WHERE VIOLENCE PERSISTS: LATENT SPATIAL FIELDS FOR THE RESIDUAL RISK OF ARMED CONFLICT AND SOCIAL UNREST

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[TODO: Abstract to be written after the application.]

1. Introduction. [TODO: Introduction (not drafted yet).]

2. Data.

2.1. *Conflict and unrest events.* Event data come from the Armed Conflict Location & Event Data Project (ACLED; Raleigh et al., 2010). ACLED records the date, location, actors and type of individual events of political violence and demonstrations, compiled from news media, local partners and other sources. We use events from 1 January 2018 to 31 December 2024 and group ACLED event types into two outcomes: *violent conflict* (battles, explosions/remote violence and violence against civilians) and *social unrest* (riots and protests). Events are aggregated to a regular $0.5^\circ \times 0.5^\circ$ longitude–latitude grid, the resolution of PRIO-GRID (Tollefsen, Strand and Buhaug, 2012), and to calendar weeks. For cell s and week t, let $N_{st}^{(j)}$ be the number of events of type j, and let

$$Y_{st} = \mathbf{1}\Big\{\textstyle\sum_{j\in\mathcal{J}} N_{st}^{(j)} > 0\Big\},$$

where $\mathcal{J}$ is the set of event types of the outcome being modelled. The two outcomes are analysed separately. Event weeks are rare. In 2023–2024, over the covered cell-weeks of Africa and the Middle East with complete history, [TODO: x]% of cell-weeks contain at least one conflict event and [TODO: x]% at least one unrest event. Over all covered cell-weeks worldwide the shares are [TODO: x]% and [TODO: x]%.

Figure 1 shows where and when events are recorded in Africa and the Middle East, the region of the main analysis. Over 2018–2024, 2.3% of covered cell-weeks in this region contain at least one conflict event and 2.3% at least one unrest event. Worldwide, the shares are 1.3% and 2.1%. Events are concentrated in space:

- a third of the cells in the region (33.6% for conflict, 35.7% for unrest) record any event at all;
- the 1% of cells with the most event weeks account for 30% of all conflict event weeks and 27% of all unrest event weeks;
- fewer than 1% of cells record events in more than half of all weeks. These cells, for example in Syria, Yemen, northeastern Nigeria and eastern Democratic Republic of the Congo, are one reason why recent history is such a strong predictor (Section 2.2).

The two outcomes have different geographies. Conflict forms large contiguous regions: the central Sahel and Lake Chad basin, the Sudans and Ethiopia, Somalia, eastern Congo, Yemen and Syria–Iraq. Unrest is densest in Mediterranean Europe, North Africa, Turkey, Iran, South

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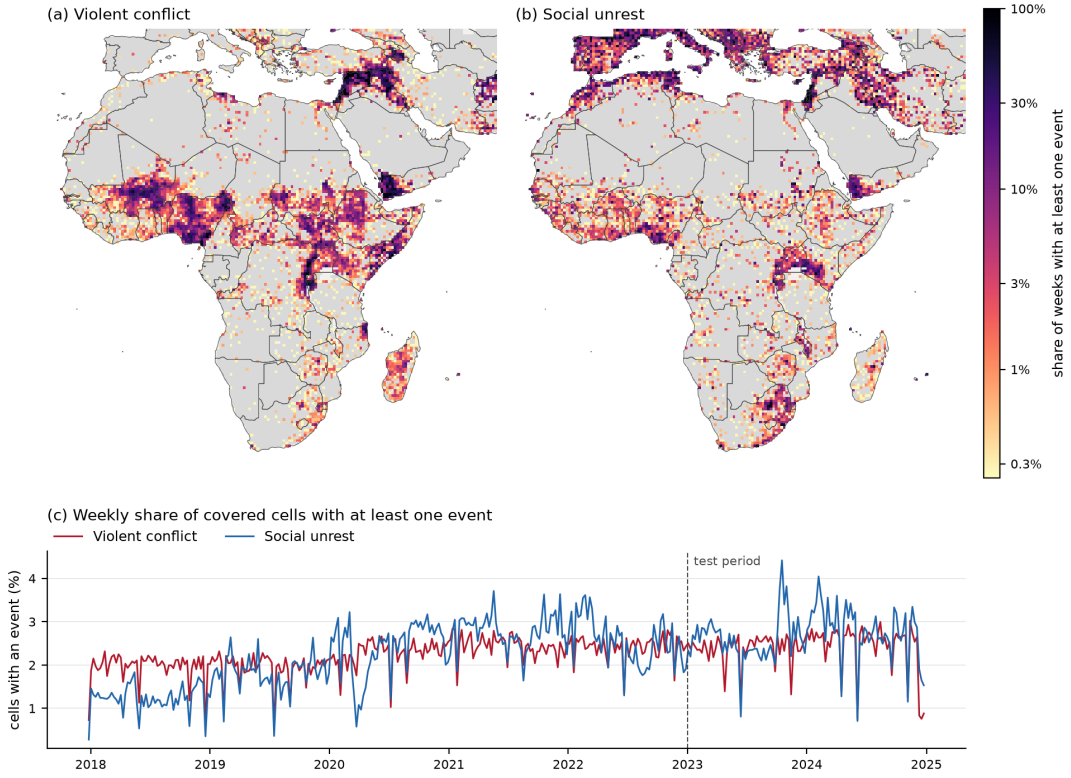


FIG 1. *ACLED events on the* 0.5° *grid in Africa and the Middle East, 2018–2024, over covered cell-weeks. (a, b) Share of weeks with at least one event in each cell, for violent conflict and social unrest (logarithmic colour scale); grey cells have no event week. (c) Weekly share of covered cells in the region with at least one event. The dashed line marks the start of the test period.*

Africa and the Great Lakes region, where population and media coverage are high. The weekly share of cells with an event rises over the period. For conflict it goes from about 2% in 2018 to about 2.5% in 2024, and for unrest from about 1.3% to between 2% and 4%. The test period is therefore not exchangeable with the early training years. The series also has isolated weeks with far fewer events than the neighbouring weeks. [TODO: Explain these weeks once the check of partition boundaries is complete; if they are an artefact of the event table, correct them.]

The grid consists of all 0.5° cells that intersect a country polygon of Natural Earth (1:10m), excluding cells without a terrestrial biome (rock and ice, and large lakes). Each cell is assigned to the country that contains its centroid. A coastal cell whose centroid falls in the sea is assigned to the nearest country. National indicators are attached through that country only, so every cell-week appears exactly once. The result is 59,344 cells observed for up to 366 weeks.

The table built by the original pipeline differs in one important respect. About 8.5% of its 34.7 million distinct cell-weeks (before the biome exclusion) occur more than once, with multiplicities such as 2^k, 3^k and $2 \cdot 3^k$, the signature of many-to-many joins between the grid and polygons of countries, IPC regions and national indicators. These copies make up 50.5% of its 70.0 million rows. The copies of a cell-week share its events but often disagree on the covariates. Among duplicated cell-weeks, the copies differ in the EPI and the liberal democracy index in 62% of cases, in GDP in 49%, in the IPC phases in about 25% and in the climate variables in 7–11%. We do not know why the climate values of a cell differ between

copies; the original pipeline does not record which polygon each copy comes from. We keep the whole first row of each cell-week, so that all covariates of a row come from the same copy. IPC phase populations are the exception: they are averaged over the copies. National indicators are re-attached for each country and year from the cells of that country that were not duplicated. GDP per capita is a national quantity: World Bank GDP of the country divided by the sum of the GHSL population of its cells. Section 5.1 describes the consequences of the duplication for the original analysis.

Two features of ACLED matter for modelling. First, geographic coverage expanded over time, and several countries entered the dataset after 2018. Cell-weeks in a country before its coverage start are *unobserved*, not event free, so they are removed rather than coded as zeros. Coverage start dates are taken from ACLED's country coverage table. Territories not listed there take the date of the country or region that contains them. Before the biome exclusion, coverage removes 7,078,822 cell-weeks in 79 countries and territories; 3,998,319 of them are in Antarctica, which the biome exclusion would remove in any case. The event history is treated as incomplete in the first four covered weeks. After both exclusions the table has 19,365,684 cell-weeks. Second, the probability that an event is recorded depends on media and partner density, so the outcome combines occurrence and reporting (Weidmann, 2016). The two are not separately identified without auxiliary information. The latent fields of Section 3 absorb spatially smooth reporting differences together with unmeasured drivers of risk, and we interpret them accordingly.

ACLED also codes the spatial and temporal precision of each event. An event whose exact location is unknown is placed at the centroid of the town or administrative unit named in the source, and an event whose exact date is unknown at a nominal date within the reported period. The data available to us are weekly counts by cell, and they carry no precision codes, so we cannot filter or weight events by precision. Imprecisely located events accumulate in the cells that contain administrative centroids and capitals. This can inflate both the persistence of events in those cells and the cell-level latent effects estimated there, and it is one more reason to read those effects as recorded rather than actual risk.

2.2. *Recent event history.* For each cell and week we compute the number of events of each type in the four preceding weeks, $L_{st}^{(j)} = \sum_{l=1}^{4} N_{s,t-l}^{(j)}$, using only past weeks of the same cell. In the models of Section 3 the history enters through $h_{st} = \log\bigl(1 + \sum_{j \in \mathcal{J}} L_{st}^{(j)}\bigr)$ for the outcome being modelled, and through the corresponding quantity for the other outcome as an additional covariate. Cell-weeks whose four-week history is incomplete, at the start of the series and of each country's coverage, are dropped from the parametric latent models. The boosting models, and the boosting offset of the hybrid models, keep these rows and receive the incomplete history as missing.

2.3. *Environmental, socioeconomic and national covariates.* Covariates are listed in Table 1. They vary at three resolutions, which is central to the identification arguments of Section 3.6:

1. *cell–week*: daily minimum and maximum temperature, heat index and wet-bulb globe temperature (CHIRTS), precipitation (CHIRPS), $PM_{2.5}$ (CAMS) and a drought index (Global Drought Observatory);
2. *cell*: population (GHSL), river length and lake area (Natural Earth), malaria risk (Malaria Atlas Project), terrestrial biome (RESOLVE Ecoregions) and distance from the equator;
3. *country–year*: GDP per capita, access to electricity, the Environmental Performance Index (EPI), the V-Dem liberal democracy index, and per-capita greenhouse gas emissions, material footprint and ecological footprint.

Food insecurity, the population in each IPC phase, is available for subnational regions and periods and is apportioned to cells by population share.

The climate variables enter all models only through their means over the four weeks before the current week, so that they precede the outcome; values of the current week are not used. The original analysis also used the current week's values. The boosting models use the four-week means on their original scale. A raw temperature means different things in different climates. In the latent models, the five climate variables (maximum and minimum temperature, precipitation, drought and $PM_{2.5}$) therefore enter as standardised anomalies with respect to a cell- and calendar-month-specific climatology estimated from the training years 2018–2022,

$$z_{st} = \big(x_{st} - \bar{x}_{s,\mathrm{m}(t)}\big)/\mathrm{sd}_{s,\mathrm{m}(t)},$$

where $\mathrm{m}(t)$ is the calendar month of week t. Precipitation is transformed to $\log(1+x)$ before the anomaly is computed. In dry seasons of arid cells the training-period standard deviation can be zero or close to zero, and any later rain would then give an extreme anomaly. We therefore floor $\mathrm{sd}_{s,\mathrm{m}}$ at the 10th percentile of the positive cell-month standard deviations of the same variable, and set $z_{st} = 0$ where the climatological standard deviation is zero. The floor applies to [TODO: x]% of the cell-months in Africa and the Middle East. The anomaly coefficients are identified from variation over time within cells. They are therefore not confounded by the time-invariant spatial pattern of the climatology. They can still be confounded by regional shocks that vary over time, since the year effects are global. The cell's mean over the training years enters separately as a cell-level covariate. A five-year climatology is noisy, especially for precipitation, and a longer reference period would be preferable where available. The anomalies are therefore best read as departures from recent conditions.

Missing covariates.. Table 1 hides substantial missingness. In the repaired table over Africa and the Middle East (4,976,634 cell-weeks), the IPC phase populations are missing in 62.7% of rows, $PM_{2.5}$ in 55.5%, and per-capita greenhouse gas emissions, material footprint and ecological footprint in 27.9%. The latter are missing for whole countries, most of them in the Democratic Republic of the Congo, Iran, Libya, Saudi Arabia, Turkey, South Sudan and Sudan. GDP is missing in 3.6% of rows, mainly in South Sudan, Yemen, Somaliland, Eritrea, Western Sahara and Syria. The EPI and the liberal democracy index are missing in 0.4% of rows, the climate means over the previous four weeks in 1.1%, and the event histories in 1.1%. Over all cell-weeks worldwide the IPC phases are missing in 89.6% of rows and precipitation in 20.8%. IPC analyses are carried out mainly where food crises are expected, and several of the countries without footprint data are among the most conflict-affected in the region. Missingness is therefore unlikely to be at random with respect to the outcome.

We keep missing values as missing through the construction of the features. In particular, a cell-week without an IPC analysis has a missing IPC value, not a food-insecure population of zero. All covariates of the latent models other than the event histories are standardised with training-period means and standard deviations. Missing values are then set to zero, the training mean, and a missingness indicator enters the model for each covariate group: one for each national indicator and one for IPC (Section 3.1). The boosting models receive the missing values as such.

2.4. *Exploratory evidence of residual spatial dependence.* Before introducing the latent models, we check whether a flexible model of the covariates and recent history leaves spatial structure in its residuals. We take the corrected XGBoost model of Section 5.2, trained on all locations, and compute the standardised cell residuals of Section 3.9 over the test period 2023–2024. In Africa and the Middle East, Moran's I of these residuals is [TODO: x] for

TABLE 1
Covariates, resolution and source.

Covariate	Resolution	Source
Minimum / maximum temperature	cell–week	CHIRTS
Heat index, wet-bulb globe temperature	cell–week	CHIRTS
Precipitation	cell–week	CHIRPS
$PM_{2.5}$	cell–week	CAMS
Drought index	cell–week	Global Drought Observatory
Events of each type in previous 4 weeks	cell–week	ACLED
Population	cell	GHSL
River length, lake area	cell	Natural Earth
Malaria risk	cell	Malaria Atlas Project
Biome	cell	RESOLVE Ecoregions
Distance from the equator	cell	derived
Food insecurity (IPC phases 1–5)	subnational–period	IPC
GDP per capita	country–year	World Bank
Access to electricity	country–year	World Bank
Environmental Performance Index	country–year	Yale EPI
Liberal democracy index	country–year	V-Dem
GHG emissions, material and ecological footprint p.c.	country–year	OWID, GFN

conflict and [TODO: x] for unrest. The residual correlation between neighbouring cells is [TODO: x] within countries and [TODO: x] across borders for conflict, and [TODO: x] and [TODO: x] for unrest. Table S7 of the Supplementary Material gives the same diagnostics for every model, and the semivariogram is reported in Section 5.3.

For the XGBoost offset cross-fitted by country (Section 3.7), the gap between the within-country and the across-border correlation is not by itself evidence of country-level effects. Neighbours in the same country are always predicted by the same fold model, which never saw that country, while neighbours across a border are usually predicted by different fold models. This design creates a country-level residual component. We therefore base the motivation for the latent effects on the corrected model, which has no such component, and report the cross-fitted offset's across-border pairs separately for countries in the same fold and in different folds ([TODO: x] and [TODO: x] for conflict). Residual structure that remains after the corrected model motivates the latent effects of Section 3.

3. Methods.

3.1. *A latent Gaussian model for event occurrence.* Let $\mathcal{S}$ be the set of n grid cells, $c(s)$ the country and $b(s)$ the biome of cell s, and $t = 1, \dots, T$ the weeks. Conditionally on a set of latent effects, the outcomes are independent Bernoulli variables, $Y_{st} \mid \pi_{st} \sim \text{Bernoulli}(\pi_{st})$, with

$$\text{logit}\, \pi_{st} = \boldsymbol{x}_{st}^{\top}\boldsymbol{\beta} + \delta\, h_{st} + \sum_{k \in \mathcal{K}} z_{k,st}\, \beta_k(s) + \boldsymbol{z}_{c(s),t}^{\top}\boldsymbol{\alpha} + u_{c(s)} + w(s) + e_s + v(s,t). \tag{3.1}$$

Here h_{st} is the recent event history of Section 2.2, so δ measures state dependence (Section 3.4). The vector $\boldsymbol{x}_{st}$ holds the cell and cell-week covariates with fixed effects. In the application these are the other outcome's event history, population, malaria risk, the IPC phase 3+ population, river length and lake area, the climate anomalies and climatologies of Section 2.3, biome effects and year effects. The variables $z_{k,st}$, $k \in \mathcal{K}$, are climate anomalies whose effects $\beta_k(s)$ vary in space (Section 3.5); in the application $\mathcal{K}$ contains maximum

temperature only. The vector $\boldsymbol{z}_{ct}$ holds the national indicators with effects $\boldsymbol{\alpha}$ (Section 3.6). GDP per capita is one of them: it is national GDP divided by national population and enters as its logarithm. The country effects $u_c \sim \mathcal{N}(0, \sigma_u^2)$ are independent, w is a Matérn field on the sphere (Section 3.2), the cell effects $e_s \sim \mathcal{N}(0, \sigma_e^2)$ are independent, and v is a space-time field (Section 3.3). The four latent terms capture persistent *recorded* risk at four scales. Each mixes occurrence with reporting (Section 2.1): national institutions and reporting regimes (u), regional hotspots that cross borders (w), single cells such as a town, a mine or a border crossing (e), and slowly evolving episodes (v). The cell effect is needed because a field with $\nu = 1$ and a range of several cells cannot represent risk that differs between adjacent cells. Without it, such risk would be absorbed by w or by the history term δh_{st} (Section 3.4).

Missing covariate values are kept as missing through the construction of the features. In particular, a cell-week without an IPC assessment has a missing IPC value, not a food-insecure population of zero. After standardisation, missing values are set to the training-period mean, and a missingness indicator is added as a further fixed effect for each covariate group: one indicator for each national indicator and one for IPC. Without these indicators, the imputed value would be confounded with the fact of not being assessed, which is itself associated with conflict. The boosting models receive the missing values as such, since XGBoost handles them natively.

Model (3.1) is a latent Gaussian model (Rue, Martino and Chopin, 2009). Stacking all latent quantities into $\boldsymbol{\xi} = (\boldsymbol{\beta}, \delta, \boldsymbol{\gamma}, \boldsymbol{\alpha}, \boldsymbol{u}, \boldsymbol{w}, \boldsymbol{e}, \boldsymbol{\eta}, \boldsymbol{v})$, the linear predictor is $\boldsymbol{\eta} = \boldsymbol{o} + \boldsymbol{A}\boldsymbol{\xi}$ for a sparse design matrix $\boldsymbol{A}$ and an optional offset $\boldsymbol{o}$, and $\boldsymbol{\xi} \mid \boldsymbol{\theta} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}(\boldsymbol{\theta})^{-1}\}$ with a sparse, block-diagonal precision whose blocks are listed in Table S1 of the Supplementary Material (Rue and Held, 2005). Fixed effects receive vague $\mathcal{N}(0, 10^4)$ priors and are integrated out together with the random effects.

3.2. *Matérn fields on the sphere via a finite-volume SPDE.* Following Lindgren, Rue and Lindström (2011), a Gaussian field w on $\mathbb{S}^2$ with Matérn-like covariance is defined as the stationary solution of

$$\tau\,(\kappa^2 - \Delta_{\mathbb{S}^2})\,w(s) = \mathcal{W}(s), \qquad s \in \mathbb{S}^2, \tag{3.2}$$

where $\Delta_{\mathbb{S}^2}$ is the Laplace–Beltrami operator and $\mathcal{W}$ is Gaussian white noise (see also Lindgren, Bolin and Rue, 2022). This is the $\alpha = 2$ case, which corresponds to Matérn smoothness $\nu = 1$ in the planar limit. The field then has practical range $\rho = \sqrt{8}/\kappa$ and marginal variance $\sigma^2 = 1/(4\pi\kappa^2\tau^2)$, provided ρ is small relative to the Earth's radius R. We parameterise the field by (ρ, σ), in kilometres and logit units.

The grid cells are natural control volumes, so instead of a triangulated mesh we discretise (3.2) by finite volumes on the longitude–latitude lattice itself (for finite-volume SPDE discretisations on regular grids, see Fuglstad et al., 2015). Integrating (3.2) over cell A_i and applying the divergence theorem with a two-point flux approximation gives

$$\tau\Big\{\kappa^2 a_i w_i + \sum_{j \sim i} \frac{\ell_{ij}}{d_{ij}}(w_i - w_j)\Big\} = \mathcal{W}(A_i) \sim \mathcal{N}(0, a_i),$$

where a_i is the area of cell i, $j \sim i$ ranges over the cells sharing an edge with i, ℓ_{ij} is the length of the shared edge and d_{ij} the great-circle distance between the centres. For a grid with spacings $\Delta\lambda, \Delta\varphi$ and centre latitude φ_i,

$$\begin{aligned} a_i &= R^2 \Delta\lambda\,\{\sin(\varphi_i + \Delta\varphi/2) - \sin(\varphi_i - \Delta\varphi/2)\}, \\ \ell_{ij}/d_{ij} &= \Delta\varphi/(\cos\varphi_i\,\Delta\lambda) \quad \text{(east–west neighbours)}, \\ \ell_{ij}/d_{ij} &= \cos\varphi_{ij}\,\Delta\lambda/\Delta\varphi \quad \text{(north–south neighbours)}, \end{aligned}$$

with φ_{ij} the latitude of the shared edge. Writing $\boldsymbol{C} = \mathrm{diag}(a_i)$ and $\boldsymbol{G}$ for the weighted graph Laplacian with edge weights ℓ_{ij}/d_{ij}, we get $\tau\boldsymbol{K}\boldsymbol{w} = \boldsymbol{C}^{1/2}\boldsymbol{\epsilon}$ with $\boldsymbol{K} = \kappa^2\boldsymbol{C} + \boldsymbol{G}$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, so

(3.3)
$$\boldsymbol{w} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}_s^{-1}\}, \qquad \boldsymbol{Q}_s(\rho, \sigma) = \tau^2\boldsymbol{K}\boldsymbol{C}^{-1}\boldsymbol{K}, \qquad \log|\boldsymbol{Q}_s| = 2n\log\tau + 2\log|\boldsymbol{K}| - \textstyle\sum_i \log a_i.$$

$\boldsymbol{Q}_s$ has at most 13 non-zeros per row. The construction is the lumped-mass finite-volume analogue of the finite-element representation in Lindgren, Rue and Lindström (2011). Because it uses the physical cell geometry, it accounts for the shrinking east–west spacing towards the poles. It can also be restricted to land cells by deleting rows and columns, and it wraps around in longitude for global analyses. The continuum relations between (κ, τ) and (ρ, σ) hold on the lattice only when ρ spans several cells. A fitted range below two or three cells (100–150 km) indicates a field that the 0.5° lattice does not resolve; its $\hat{\sigma}$ is then not the marginal standard deviation, and we interpret such fields as cell-level effects.

REMARK 1. The discretisation imposes reflecting (Neumann) conditions on the boundary of the lattice, which roughly doubles the marginal variance of boundary cells (Section 4.2; see Khristenko et al., 2019, for the effect of boundary conditions on SPDE fields). For a land grid the boundaries are the coastlines. Extending the lattice into the sea removes the inflation but lets correlation leak across narrow seas, for example between Yemen and Somalia across the Gulf of Aden, which is substantively wrong for conflict risk. We therefore use the barrier model of Bakka et al. (2019): the SPDE is written as

$$u(s) - \nabla \cdot \left\{ \tfrac{r(s)^2}{8} \nabla u(s) \right\} = r(s)\sqrt{\pi/2}\,\sigma\,\mathcal{W}(s),$$

which reduces to (3.2) with $\kappa = \sqrt{8}/\rho$ when $r(s) \equiv \rho$. The lattice is extended by six rings of cells (3°) without data. Sea cells, identified from the Natural Earth country polygons, get range $r(s) = f\rho$. Land cells without data, such as those outside a regional analysis box, keep the range ρ, so that the field is not artificially decorrelated at the edge of the region. The finite-volume form is $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{D}\boldsymbol{\epsilon}$ with $\boldsymbol{K} = \boldsymbol{C} + (\rho^2/8)\boldsymbol{G}_m$, where $\boldsymbol{G}_m$ has edge weights $2m_im_j/(m_i+m_j)\cdot\ell_{ij}/d_{ij}$, $m_i = r_i^2/\rho^2$ (the harmonic mean, the usual two-point flux choice for a discontinuous coefficient), and $\boldsymbol{D} = \mathrm{diag}\{r_i\sigma(\pi a_i/2)^{1/2}\}$, so that $\boldsymbol{Q}_s = \boldsymbol{K}\boldsymbol{D}^{-2}\boldsymbol{K}$. Table S2 of the Supplementary Material reports the marginal variance at the coast and the correlation across straits of two to six cells for several choices of f, at $\rho = 400$ km. At this range, $f = 0.1$, the value of Bakka et al. (2019), gives a barrier range of 40 km, less than one cell, and the coastline behaves almost like a reflecting boundary. We use $f = 0.5$. At $\rho = 400$ km it keeps the coastal variance within about 30% of nominal and the correlation across straits of four cells or more (about 220 km) below 0.04, while a strait of two cells still has a correlation of 0.16. Straits narrower than one cell are not resolved at this grid resolution. These figures depend on ρ, because the barrier range $f\rho$ and the buffer of six rings scale differently with it. For the fitted ranges of w in the application, which are well below 400 km, the barrier is stronger than in Table S2 of the Supplementary Material. For ranges of 800 km or more, which the SVC fields of Section 3.5 can reach, $f\rho$ exceeds the buffer, so the coastal variance inflation returns and correlation leaks across the wider straits. The stored real-data fits use the sea cells of the country polygons as barriers.

Priors.. We use penalised-complexity priors (Simpson et al., 2017; Fuglstad et al., 2019): $P(\rho < \rho_0) = 0.05$ and $P(\sigma > \sigma_0) = 0.01$, with $\rho_0 = 100$ km (about two grid cells) and $\sigma_0 = 1$. The PC prior of Fuglstad et al. (2019) is derived for Matérn fields on $\mathbb{R}^2$; we use it as an approximation on the sphere, which is adequate for ranges far below R. The country and cell effects have PC priors with $P(\sigma_u > 1) = 0.01$ and $P(\sigma_e > 1) = 0.01$.

3.3. *Space-time persistence.* The space-time term is a separable field that evolves as a first-order autoregression of Matérn innovations,

$$\boldsymbol{v}_\tau = \phi\, \boldsymbol{v}_{\tau-1} + \sqrt{1-\phi^2}\, \boldsymbol{\epsilon}_\tau, \qquad \boldsymbol{\epsilon}_\tau \sim \mathcal{N}\{\mathbf{0}, \boldsymbol{Q}_s(\rho_v, \sigma_v)^{-1}\},$$

(Cameletti et al., 2013), so that $\boldsymbol{v}$ has precision $\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$, where $\boldsymbol{Q}_{\mathrm{AR}}$ is the tridiagonal precision of a unit-variance AR(1) process. To control cost, τ indexes blocks of Δ weeks, and $v(s,t) = v_{s,\lceil t/\Delta \rceil}$. We use $\Delta = 1$ and $\Delta = 4$ in the simulations and $\Delta = 8$ in the application. The field v is defined on the observed land cells only, so it cannot link cells across the sea, and it has reflecting boundaries at the edges of the observed region. In the models with v, the static field w uses the same lattice as elsewhere: the observed cells, a buffer of six rings and sea barriers with $f = 0.5$ (Remark 1). The prior on $\operatorname{atanh}\phi$ is $\mathcal{N}(0, 1.5^2)$. Non-separable alternatives exist (Lindgren et al., 2024) but are not needed for the questions asked here.

3.4. *State dependence versus persistent heterogeneity.* Recent event counts dominated the importance rankings of the original boosting analysis (Menicali, Marcantonio and Castruccio, 2026). Part of that dominance was an artefact of duplicated rows (Section 5.2), but after correction recent history remains the strongest single predictor. This has several explanations, which model (3.1) separates only partially:

1. *state dependence*: recorded events raise the probability of further recorded events in the same place, through retaliation, mobilisation or the arrival of armed groups;
2. *persistent heterogeneity*: some places are simply riskier (u, w, e or v is large there), and past events are informative only as a proxy for that latent risk (Heckman, 1981);
3. *reporting dynamics*: media attention and reporting networks themselves persist, so a place that has been reported on is more likely to be reported on again, whatever happens there (Weidmann, 2016).

All three produce strong autocorrelation in Y_{st} and a large predictive contribution of h_{st}. In the conflict literature the distinction appears as the question of whether apparent contagion between neighbouring places reflects spillover or shared conditions (Buhaug and Gleditsch, 2008). Self-exciting point process models address it by decomposing intensity into a background and a triggering component (Mohler et al., 2011; Mohler, 2013; Porter and White, 2012; Clark and Dixon, 2018; Reinhart, 2018; Jun and Cook, 2024).

In model (3.1), δ is the effect of recent recorded history conditional on the latent persistence $u + w + e + v$, and it is identified only under the assumed latent structure. It is conditional on heterogeneity at the scales that are modelled. Persistent heterogeneity that none of the latent terms can represent loads onto δh_{st} and inflates $\hat{\delta}$. This is why the cell effect e is included: without it, time-invariant risk that varies between adjacent cells could only be represented by h_{st}. The space-time field v is resolved at steps of Δ weeks, and h_{st} covers the previous four weeks. When the two time scales are similar, separating δ from v relies on the AR(1) form, so we report δ for several Δ (Section 4.5). We therefore read $\hat{\delta}$ as conditional dependence of recorded events, not as a causal contagion effect. To limit the initial-conditions problem of dynamic random-effects models (Wooldridge, 2005), we drop weeks whose four-week history is incomplete (the first weeks of each cell and of each country's ACLED coverage). We also add each cell's history at its first usable week as a covariate, following Wooldridge (2005). This is a limited version of that device. A single four-week window is a noisy proxy for a cell's initial state, and its coefficient does not vary in time, whereas under the AR(1) form of v the dependence of later weeks on the initial state decays with time. The device therefore only partly removes the initial-conditions bias.

3.5. *Spatially varying climate effects.* To assess whether climate effects differ across biomes and how much variation remains within them, the effect of anomaly k is decomposed as

$$\beta_k(s) = \gamma_{k,b(s)} + \eta_k(s), \qquad \boldsymbol{\eta}_k \sim \mathcal{N}\{\mathbf{0}, \boldsymbol{Q}_s(\rho_k, \sigma_k)^{-1}\}, \tag{3.4}$$

a spatially varying coefficient model (Gelfand et al., 2003; Dambon, Sigrist and Furrer, 2021) whose mean is pooled by biome. The biome means $\gamma_{k,b}$ are fixed effects with the vague prior of Section 3.1. The field $\boldsymbol{\eta}_k$ has PC priors with $P(\rho_k < 100\,\mathrm{km}) = 0.05$ and $P(\sigma_k > 0.5) = 0.01$, and σ_k measures heterogeneity within biomes. The design of $\boldsymbol{\eta}_k$ is the sparse matrix with entry $z_{k,st}$ in row (s,t) and column s.

Three summaries of a biome-specific effect should be distinguished:

- *Fixed biome slopes.* A model with fixed biome-specific slopes and no $\boldsymbol{\eta}_k$ estimates an information-weighted average, $\sum_{s\in\mathcal{S}_b} \omega_s \beta_k(s) / \sum_{s\in\mathcal{S}_b} \omega_s$, with $\omega_s \approx \sum_t z_{k,st}^2 \pi_{st}(1-\pi_{st})$. This average puts most weight on the high-risk cells of the biome.
- *Biome means.* Model (3.4) targets the biome mean $\gamma_{k,b}$, the mean of the process that generated the biome's coefficients. This is a superpopulation quantity. Because $\boldsymbol{\eta}_k$ is not constrained to average zero within biomes and biomes are large contiguous regions, $\gamma_{k,b}$ is separated from the biome average of $\boldsymbol{\eta}_k$ mainly when ρ_k is small relative to the size of the biome. When ρ_k is comparable to biome size, the separation rests partly on the prior. A nearly constant component of $\boldsymbol{\eta}_k$ is then confounded with a common shift of all $\gamma_{k,b}$. The intervals for $\gamma_{k,b}$ are conditional on $(\hat{\rho}_k, \hat{\sigma}_k)$ (Section 3.8), and in this situation the uncertainty in these hyperparameters may be the largest source of error. They should be read with this in mind.
- *Median SHAP value.* The summary used in the SHAP-based analysis of Menicali, Marcantonio and Castruccio (2026), the median SHAP value within a biome, differs from both. The following identities are on the logit scale of the model. They hold for interventional SHAP values when the features are independent. The path-dependent TreeSHAP of XGBoost (Lundberg et al., 2020), which we use, satisfies them only approximately, and features correlated with biome, such as distance from the equator, can absorb part of an interaction. For a feature with an additive effect $f_k(z)$, the SHAP value is $f_k(z) - \mathbb{E}f_k(Z_k)$, so the median within a biome is $\mathrm{median}_{(s,t):\,b(s)=b}\{f_k(z_{k,st}) - \mathbb{E}f_k(Z_k)\}$. For a linear effect $\beta_k z$ this is β_k times the median of $z_{k,st} - \mathbb{E}z_k$ in the biome. It reflects where the biome's covariate values lie relative to the global mean as much as the strength of the effect. When the effect is a biome-by-feature interaction, SHAP also splits the interaction between the feature and biome (Kumar et al., 2020; Aas, Jullum and Løland, 2021). A pure interaction $g(b)z$, with B independent of Z, contributes $\{g(b) + \mathbb{E}g(B)\}(z - \mathbb{E}z)/2$ to the SHAP value of z. Within biome b, the least-squares slope of the SHAP value of z on z therefore targets $\{g(b) + \mathbb{E}g(B)\}/2$, which is pulled towards the average effect across biomes. It equals $g(b)/2$ only when $\mathbb{E}g(B) = 0$. The SHAP value with the interaction credited fully to z, $\Phi_{zz} + 2\sum_{j\neq z}\Phi_{zj}$ with Φ the SHAP interaction values (Lundberg et al., 2020), equals $g(b)(z - \mathbb{E}z)$ in this case, so its slope within biome b targets $g(b)$ whatever the mean of g.

3.6. *National indicators and spatial confounding.* National indicators $\boldsymbol{z}_{ct}$, such as GDP per capita, the EPI or the ecological footprint, take one value per country and year. GDP per capita is computed at the national level (Section 3.1), so that it is constant within each country and year like the other indicators. We decompose each into a between-country component, its country mean over the training years $\bar{z}_c$, and a within-country component $z_{ct} - \bar{z}_c$, with separate coefficients (a Mundlak decomposition). Year effects absorb trends common to all

countries. The between coefficients are identified from contrasts between the n_c countries only, so their effective sample size is of order n_c, not the number of cell-weeks. Modelling cell-weeks as conditionally independent, or bootstrapping spatial blocks that cut across countries, therefore produces intervals that are far too narrow. The within coefficients rest on the few years of change within each country. A country with no training-period value of an indicator gets the mean between component and a value of one for that indicator's missingness indicator (Section 3.1). The missingness indicator is itself country-level, so its coefficient also rests on few contrasts.

If the indicators are associated with unobserved country-level risk, so that $\mathbb{E}\{u_c + \bar{w}_c \mid \bar{z}_c\} \neq 0$ with $\bar{w}_c$ the country average of w, then the between coefficients are confounded. An indicator of environmental performance that tracks state capacity, or reporting capacity, is the leading example. This is the spatial confounding problem (Reich, Hodges and Zadnik, 2006; Hodges and Reich, 2010; Paciorek, 2010). Adding a spatial random effect changes the estimate of a spatially smooth covariate's coefficient, and in general neither the estimate with the random effect nor the one without it is unbiased for a causal effect (Khan and Calder, 2022; Zimmerman and Ver Hoef, 2022). Recent work has made this more precise. Generalised least squares with a spatial random effect can be consistent when the covariate has non-spatial variation (Gilbert, Ogburn and Datta, 2025). Confounding in the data-generating process should be distinguished from the behaviour of the analysis model (Khan and Berrett, 2026). The bias depends on how the covariate and the confounder are spread across spatial scales (Guan et al., 2023; Dupont, Marques and Kneib, 2026). A country-constant indicator has no within-country spatial variation, so these results give little reassurance here.

We report the following estimators and treat their disagreement as a measure of how fragile the national-level conclusions are:

1. model (3.1) without latent effects;
2. model (3.1) with $\boldsymbol{u}$ and $\boldsymbol{e}$;
3. model (3.1) with $\boldsymbol{u}$, $\boldsymbol{w}$ and $\boldsymbol{e}$;
4. a *spatial+* estimator (Dupont, Wood and Augustin, 2022): model (3.1) with $\boldsymbol{w}$ and $\boldsymbol{e}$ but *without* $\boldsymbol{u}$, in which each between component is replaced by its residual after an area-weighted projection, over the observed land cells, on the m smoothest eigenvectors $\boldsymbol{\psi}$ of $\boldsymbol{G\psi} = \lambda \boldsymbol{C\psi}$.

These eigenvectors are the lattice analogue of the thin-plate eigenvectors of Dupont, Wood and Augustin (2022). A country-constant indicator varies only by steps at borders. After the projection, the residual keeps these steps, minus a smooth function of the eigenvectors. In a model without $\boldsymbol{u}$, the smooth variation is left to $\boldsymbol{w}$ and the border steps identify the coefficient. In a model with $\boldsymbol{u}$, the border steps are country-level jumps, which $\boldsymbol{u}$ can absorb. The coefficient would then be identified only through the shrinkage of $\boldsymbol{u}$ under its exchangeable prior, which is independent of the indicator. That is the random-effects exogeneity assumption we have just called implausible. This is why (iv) omits $\boldsymbol{u}$. Without $\boldsymbol{u}$, however, country-level risk that is not spatially smooth remains a possible confounder, and the projection also gives the residual some within-country variation that is a function of the smooth eigenvectors. We therefore treat (iv) as a sensitivity analysis and not as a preferred estimate.

The default m^* is the number of eigenvalues below $\kappa_w^2 = 8/\hat{\rho}_w^2$, the eigenvectors that vary on scales longer than the fitted range of w. This number can be large. By Weyl's law, for a region of area $|D|$ it is approximately $|D|\kappa_w^2/(4\pi) = 2|D|/(\pi\hat{\rho}_w^2)$. If $\hat{\rho}_w$ is close to the ranges of the hybrid fields of Section 5.3 (264 km for conflict and 68 km for unrest, on 13,719 cells), m^* is of the order of a few hundred for conflict and several thousand for unrest. The eigenvectors near the cut-off then vary on scales comparable to the size of many countries, and most of the between-country variation of a country-constant indicator is projected out.

We therefore report $m = 10$, $m = 20$ and m^*, and state m^* for each fit [TODO: values of m^* from the rerun of fit_glmm.py, and whether the eigenvalue computation reached m^*]. The restricted spatial regression of Reich, Hodges and Zadnik (2006) is not used, because its intervals are known to be anticonservative (Hanks et al., 2015; Khan and Calder, 2022).

3.7. *Hybrid models: boosting offsets with latent effects.* To retain the flexibility of gradient boosting for the many weakly informative covariates, we also fit a semiparametric version of (3.1). Its fixed part is replaced by the logit $\hat{f}(\boldsymbol{x}_{st})$ of an XGBoost classifier (Chen and Guestrin, 2016), which enters as an offset:

$$\operatorname{logit} \pi_{st} = \hat{f}^{(-k(s))}(\boldsymbol{x}_{st}) + \beta_0 + u_{c(s)} + \omega(s), \tag{3.5}$$

where ω is either the Matérn field w or independent cell effects. The offset is *cross-fitted by country*. Countries are assigned at random to $K = 5$ folds. For each fold k, $\hat{f}^{(-k)}$ is trained on the 2018–2022 rows of the other folds, with its number of trees chosen by early stopping on 2022 after training on 2018–2021. It then provides the offset for every row of fold k, in both the training and test periods. The offset and the latent effects are fitted on the Africa and Middle East box only. Trees that receive national indicators can identify countries, and through population, biome and distance from the equator, approximate locations. Cross-fitting by country ensures that the offset of a cell was learned without any data from its own country, so that u does not count that country's data twice. It does not exclude data from neighbouring countries. The offset of a cell near a border can be learned from the outcomes of cells across the border, and ω is estimated from the residuals of those same cells. Regional hotspots that cross borders can therefore enter both the offset and ω. The fitted ω is a residual field. Its range and standard deviation are not comparable with those of w in model (3.1). The offset enters with its coefficient fixed at one, so fold models whose logits are over- or under-dispersed are not rescaled; u absorbs shifts between folds but not differences in scale. Inference for u and ω is conditional on the offset, whose estimation uncertainty is not propagated, so their posterior uncertainty is understated. Cross-fitting is used here in the loose sense of sample splitting, not with an orthogonal score as in Chernozhukov et al. (2018).

The price of cross-fitting is that the offset is weaker than a boosting model trained on all locations. The cross-fitted XGBoost model never sees the country it predicts, while u and ω are estimated from that country's training data. A gain of the hybrid model over the cross-fitted offset alone is therefore not evidence that the latent effects add anything. The hybrid models are compared, on identical test rows, with the corrected boosting model trained on all locations, with the location-aware boosting baseline, with each other (ω as a field or as independent cell effects) and with the persistence baseline (Section 3.9). The offset uses the same rows, features and hyperparameters as the corrected boosting model (Section 5.2). It is trained without class weights, so it is on the log-odds scale.

Model (3.5) differs from GPBoost (Sigrist, 2022, 2023), which alternates boosting steps with latent Gaussian model updates: here the boosting fit is fixed before the latent model is fitted. This makes the model cheap enough for the global data and allows the same Laplace machinery to be used. Random forests with spatially correlated errors (Saha, Basu and Datta, 2023) are a further alternative.

The boosting analysis of Menicali, Marcantonio and Castruccio (2026) trained XGBoost with positives weighted by $\omega_+ = (n_0/n_1)^{1/2}$. For the unrestricted minimiser of the logistic loss, this shifts the fitted log-odds by $\log \omega_+$ (King and Zeng, 2001). We therefore prior-correct the predictions of the weighted models, $\hat{f} \leftarrow \hat{f} - \log \omega_+$, before probabilistic scoring. For boosted trees the correction is only approximate, because the weights also act on the leaf penalty, on the minimum leaf weight, which is defined on weighted Hessians, and on early stopping. The scores of the weighted models therefore include any remaining miscalibration. Class reweighting does not improve discrimination and miscalibrates probabilities (van den Goorbergh et al., 2022), and the corrected boosting model is trained without it (Section 5.2).

3.8. *Inference.* For fixed hyperparameters $\boldsymbol{\theta}$, the posterior mode $\boldsymbol{\xi}^* = \arg\max_{\boldsymbol{\xi}}\{\log p(\boldsymbol{y} \mid \boldsymbol{\xi}) - \frac{1}{2}\boldsymbol{\xi}^\top \boldsymbol{Q}\boldsymbol{\xi}\}$ is found by damped Newton iterations,

$$\boldsymbol{\xi} \leftarrow \boldsymbol{\xi} + \lambda\,(\boldsymbol{Q} + \boldsymbol{A}^\top \boldsymbol{W}\boldsymbol{A})^{-1}\{\boldsymbol{A}^\top(\boldsymbol{y} - \boldsymbol{\pi}) - \boldsymbol{Q}\boldsymbol{\xi}\}, \qquad \boldsymbol{W} = \operatorname{diag}\{\pi_{st}(1-\pi_{st})\},$$

with step halving. The Laplace approximation to the marginal likelihood is

$$\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) = \log p(\boldsymbol{y} \mid \boldsymbol{\xi}^*) - \tfrac{1}{2}\boldsymbol{\xi}^{*\top}\boldsymbol{Q}\boldsymbol{\xi}^* + \tfrac{1}{2}\log|\boldsymbol{Q}| - \tfrac{1}{2}\log|\boldsymbol{Q} + \boldsymbol{H}|, \qquad \boldsymbol{H} = \boldsymbol{A}^\top \boldsymbol{W}^* \boldsymbol{A}. \tag{3.6}$$

The hyperparameters are set to $\hat{\boldsymbol{\theta}} = \arg\max\{\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})\}$ by L-BFGS-B with finite-difference gradients on the internal scale (logarithms of ranges and standard deviations, $\operatorname{atanh}\phi$). Each evaluation warm-starts Newton at the previous mode. Given $\hat{\boldsymbol{\theta}}$, the latent posterior is approximated by $\mathcal{N}\{\boldsymbol{\xi}^*, (\boldsymbol{Q} + \boldsymbol{H})^{-1}\}$. Samples reuse the factorisation of $\boldsymbol{Q} + \boldsymbol{H}$ already computed at the mode, through the perturbation identity

$$\boldsymbol{\xi} = \boldsymbol{\xi}^* + (\boldsymbol{Q} + \boldsymbol{H})^{-1}\big(\boldsymbol{R}^\top \boldsymbol{\epsilon}_1 + \boldsymbol{A}^\top \boldsymbol{W}^{*1/2}\boldsymbol{\epsilon}_2\big),$$

where $\boldsymbol{R}^\top \boldsymbol{R} = \boldsymbol{Q}$ is block-wise explicit, for instance $\boldsymbol{R} = \boldsymbol{D}^{-1}\boldsymbol{K}$ for a Matérn block and a Kronecker product of a bidiagonal AR(1) factor with that matrix for $\boldsymbol{v}$. This is an empirical Bayes version of INLA (Rue, Martino and Chopin, 2009) and is similar in spirit to TMB (Kristensen et al., 2016). Laplace approximations are generally accurate for binary GLMMs but can understate uncertainty when there are few observations per latent variable and events are rare (Fong, Rue and Wakefield, 2010; Ferkingstad and Rue, 2015). The intervals also ignore the uncertainty in $\hat{\boldsymbol{\theta}}$. Section 4.2 therefore checks coverage at the event prevalence of the data, and Section 4.5 checks it for the space-time model, where each latent value is informed by only Δ Bernoulli observations.

Computation.. All matrices are sparse. Every Newton step and every log-determinant requires a sparse factorisation of $\boldsymbol{Q} + \boldsymbol{H}$, whose sparsity pattern does not depend on $\boldsymbol{\theta}$. We use a sparse LU factorisation in symmetric mode (SuperLU). For matrices of dimension above 2,000, the fill-reducing ordering is a nested dissection computed by METIS (Karypis and Kumar, 1998); it is computed once and reused, but the factorisation itself is recomputed at every Newton step and every evaluation of (3.6). If METIS is not available, the code falls back to multiple minimum degree on $\boldsymbol{A}^\top + \boldsymbol{A}$. For the space-time model the nested-dissection ordering reduced factorisation time by a factor of 7 to 14 compared with that fallback. An LU factorisation stores both triangular factors, so it needs about twice the memory of a Cholesky factor of the same matrix. Newton iterations stop after at most 50 steps, and a value of $\boldsymbol{\theta}$ at which the factorisation fails is given a very low objective value. Both events can make the optimiser stop at a poor point, so they should be recorded for each fit [TODO: report Newton non-convergence and failed evaluations for the real-data fits]. Since $\boldsymbol{H} = \boldsymbol{A}^\top \boldsymbol{W}\boldsymbol{A}$ aggregates over weeks, the cost of the spatial and SVC blocks is driven by the number of cells, not by the number of cell-weeks. A global 0.5° land grid gives about 6×10^4 latent values per spatial field. The space-time block has nT_v latent values, which is why it is fitted on a regional subset. The fixed-effect and country columns add dense rows to $\boldsymbol{Q} + \boldsymbol{H}$; with tens of fixed effects their cost is small.

Prediction.. For a new cell-week with design row $\boldsymbol{a}$ and offset o, the posterior predictive probability is $\mathbb{E}\{\operatorname{expit}(o + \boldsymbol{a}^\top\boldsymbol{\xi})\} \approx \operatorname{expit}\{m/\sqrt{1 + \pi s^2/8}\}$, where m and s^2 are the posterior mean and variance of $o + \boldsymbol{a}^\top\boldsymbol{\xi}$; s^2 is estimated from posterior samples. Test years beyond the training period take the year effect of the last training year.

3.9. *Diagnostics and evaluation.*

Residual diagnostics.. Binary residuals are uninformative one at a time, so we aggregate over the test weeks within cells. The standardised cell residual is

$$r_s = \sum_t (y_{st} - \hat{\pi}_{st}) \Big/ \Big\{ \sum_t \hat{\pi}_{st}(1 - \hat{\pi}_{st}) \Big\}^{1/2},$$

with the sums over the test period 2023–2024. All residual diagnostics reported in the paper are computed in this way for every model, including the cross-fitted offset, on the same test rows. Residuals of in-sample latent-model fits are shrunk and spatially dependent through the smoother, and we do not use training-period residuals as evidence.

Even for a correct model, the test residuals are neither standard normal nor independent across cells. The predictions $\hat{\pi}_{st}$ carry the estimation error of the training period, for example in $\hat{w}(s)$. A logit error ϵ_s contributes about $\epsilon_s\{\sum_t \pi_{st}(1-\pi_{st})\}^{1/2}$ to r_s, which grows with the number of test weeks, and the errors of a smoothed field are positively correlated between neighbours. The diagnostics below are therefore used to compare models on the same rows, not as tests against a null of independence. We report:

- Moran's I (Moran, 1950) of $\{r_s\}$ for rook adjacency. We also give a permutation p-value from $B = 999$ permutations, which is bounded below by $1/(B+1)$. Its null hypothesis of exchangeable residuals does not hold even for a correct model, so it is descriptive only;
- the empirical semivariogram of $\{r_s\}$ against great-circle distance;
- the correlation of r_s between neighbouring cells in the same country and across a border.

A larger within-country correlation can point to effects at the country level, of occurrence or of reporting. For the cross-fitted offset it also has a mechanical source. Neighbours in the same country are always predicted by the same fold model, while neighbours across a border are usually predicted by different fold models, whose differences create a country-level residual component. For this model we therefore also split the across-border pairs into pairs whose two countries are in the same fold and pairs in different folds.

Out-of-sample evaluation.. Models are trained on 2018–2022 and evaluated on 2023–2024, matching the original analysis. All models are compared on identical test rows. We report the logarithmic score and the Brier score, both strictly proper (Gneiting and Raftery, 2007), and the area under the precision–recall curve, computed as average precision (Boyd, Eng and Page, 2013). The latter is more informative than the ROC curve at the low prevalence of the data (Saito and Rehmsmeier, 2015). Paired differences in log score and Brier score are reported with two standard errors: one clustered by cell and one clustered by country, with the number of clusters. The cell-clustered errors allow for dependence over time within a cell but not between cells. The latent fields and the national indicators induce dependence at regional and national scales (Section 3.6), so the country-clustered errors are the more reliable of the two, although they rest on few clusters. Since prediction in the same places is dominated by persistence (Mueller and Rauh, 2022), every model is compared with a persistence baseline that ranks cell-weeks by the number of events in the previous four weeks. We also report all scores on *onset* rows: test cell-weeks in cells that had no event of that outcome in the previous 52 weeks. On these rows recent history carries no information, so they show how well a model anticipates events in places that have been quiet.

We also compare every model with a *location-aware* boosting baseline that adds each cell's training-period event rate as a feature. For the fit used to choose the number of trees by early stopping on 2022, this rate is computed from 2018–2021 only, so it uses no 2022 outcomes. For the final fit on 2018–2022 it is computed leaving out the row's own year, and for the test rows it is the rate over 2018–2022. This baseline is the boosting counterpart of a cell intercept, and the latent fields have to improve on it to be worth their cost.

4. Simulation studies. The simulations address five questions that the application raises but cannot settle on its own:

(S0) whether the finite-volume SPDE and the empirical Bayes Laplace approximation recover the field and give calibrated intervals, including at the event prevalence of the data;
(S1) whether boosting leaves residual spatial structure, and whether modelling it improves forecasts for the same places beyond a location-aware boosting baseline;
(S2) how national indicators behave under spatial confounding;
(S3) whether state dependence can be separated from persistent heterogeneity;
(S4) whether biome-specific climate effects are recovered better by spatially varying coefficients than by SHAP summaries.

All code is in the supplementary material.

4.1. *Data-generating process.* The simulator mimics the structure of the ACLED grid. The domain is a 40×30 block of 0.5° cells between 10°E and 30°E and between 5°S and 10°N (1,200 cells of about 55 km), with 104 training weeks, 52 test weeks and 12 burn-in weeks. Countries are the Voronoi cells of randomly placed seeds (14 countries; 25 in S2). Four biomes are latitude bands with boundaries perturbed by a smooth random field, ordered from "cold" (biome 1) to "tropical" (biome 4). Population is a standardised Matérn field (range 250 km) plus white noise. Temperature and precipitation anomalies are standardised AR(1) processes in time (coefficients 0.6 and 0.3) of Matérn fields with ranges of 800 and 500 km. A raw temperature is also generated, as a smooth climatology that rises from biome 1 to biome 4, plus a seasonal cycle and 2.5 times the anomaly. Three national covariates Z_0, Z_1, Z_2 are standard normal across countries. Z_1 and Z_2 have no effect but, like the national indicators in the data, allow trees to identify countries. Outcomes are generated sequentially from

(4.1)
$$\operatorname{logit} \pi_{st} = \beta_0 + \beta_{\text{pop}} \text{pop}_s + \delta \log\Big(1 + \textstyle\sum_{l=1}^{4} y_{s,t-l}\Big) + \beta(s)\, \text{temp}_{st} + \beta_{\text{prec}} \text{prec}_{st} + \alpha Z_{0,c(s)} + u_{c(s)} + w(s) + v(s,t),$$

with $\beta(s) = \gamma_{b(s)} + \eta(s)$ and $u_c \sim \mathcal{N}(0, 0.3^2)$. To avoid generating data from the model used for fitting, w, η and the innovations of v are drawn from Gaussian processes with the continuous Matérn covariance ($\nu = 1$) in great-circle distance, evaluated at the cell centres. They are not drawn from the lattice GMRF of Section 3.2. Unless stated otherwise, $\beta_0 = -4$, $\beta_{\text{pop}} = 0.5$, $\delta = 0.6$, $\boldsymbol{\gamma} = (-0.3, -0.1, 0.1, 0.3)$, $\eta \equiv 0$, $\beta_{\text{prec}} = -0.15$, $\alpha = 0$, $\sigma_w = 1$, $\rho_w = 400$ km and $v \equiv 0$. The resulting prevalence of event weeks is about 4%. Spatial confounding (S2) is induced by setting $Z_{0,c} = \varrho\, \tilde{\ell}_c + (1 - \varrho^2)^{1/2} \epsilon_c$, where $\tilde{\ell}_c$ is the standardised country average of $u_c + w(s)$ and $\epsilon_c \sim \mathcal{N}(0, 1)$.

The default parameters are not calibrated to the application. A first fit of the hybrid model to the conflict outcome, before the revisions of Section 5.3, gave a field standard deviation of about 1.5, a range of about 260 km and a country standard deviation of about 0.8, and there persistence was about as accurate as cross-fitted boosting (AUC-PR 0.45 and 0.44). [TODO: Check these against the final fit in Section 5.3.] S1 is therefore repeated with $\sigma_w = 1.5$, $\rho_w = 260$ km and $\sigma_u = 0.8$, other parameters at their defaults. All covariate effects in (4.1) are linear on the logit scale, and the only interaction is between biome and temperature. This setting does not reward the flexibility of trees, so the comparisons between boosting and latent models below hold for this data-generating process and need not carry over to the application.

With R replicates, a proportion p (coverage, rejection rate) has Monte Carlo standard error $\{p(1-p)/R\}^{1/2}$, at most 0.09 for $R = 30$ and 0.11 for $R = 20$, and a mean estimate has Monte Carlo standard error equal to its empirical SD divided by $R^{1/2}$. We give these standard errors where a comparison depends on them.

Boosting.. Two versions of XGBoost are used, both with the hyperparameters of the original analysis (Menicali, Marcantonio and Castruccio, 2026): depth 5, learning rate 0.05, row and column subsampling 0.8, minimum child weight 30 and L_2 penalty 5. The *original-like* version follows the original procedure. Positives are weighted by $(n_0/n_1)^{1/2}$. The number of trees is chosen by early stopping on the last 26 training weeks, after which the model is refitted on all training weeks. Its log-odds are prior-corrected before scoring (Section 3.7). The *unweighted* version uses 300 trees without class weights. The minimum child weight of 30 was chosen for millions of rows. At a prevalence of about 4% each row contributes a Hessian of about 0.04, so a leaf needs several hundred rows, which restricts the trees more on the simulated training sets (about 125,000 rows, or 37,000 for a 30% subsample) than on the data. The features are population, distance from the equator, biome, the two anomalies, Z_0–Z_2 and the four-week event count. As in the original analysis, they include no coordinates.

4.2. *S0: accuracy of the field representation and of the Laplace engine.* We draw w on the 40×30 domain from the continuous Matérn process with $\rho_w = 400$ km and $\sigma_w = 1$, and Bernoulli outcomes from $\operatorname{logit} \pi_{st} = \beta_0 + 0.5\, x_{st} + w(s)$ with $x_{st} \sim \mathcal{N}(0,1)$, in two settings with 50 replicates each. In the *common* setting, $\beta_0 = -2$ with 40 weeks (prevalence about 15%). In the *rare* setting, $\beta_0 = -4.5$ with 104 weeks (prevalence about 2%, as in the data). We also report the marginal variance of the lattice field itself, from 2,000 exact draws, in interior cells and at the edge of the domain. Table 2 summarises the results.

The lattice field has marginal variance 1.07 in cells at least eight cells (about 440 km) from the edge of the domain, against a nominal 1, and 2.02 at the edge. This is the doubling induced by the reflecting boundary that motivates the barrier model of Remark 1. Although the data are not generated from the lattice model, the range is recovered with little bias in both settings ($\hat{\rho}_w = 420$ and 409 km against 400). The standard deviation is underestimated by about 12% ($\hat{\sigma}_w = 0.88$ against 1). The bias is the same at prevalences of 16% and 2%, so it is not mainly due to the limited information in rare binary outcomes. A plausible cause is the mismatch between the lattice model and the continuous process, including the inflated variance near the boundary, since only about 28% of the cells are at least eight cells from the edge. We have not tested this explanation. The posterior mean of w has correlation 0.95 with the truth in the common setting and 0.91 in the rare one. Plug-in 95% intervals cover b_1 in 88% of replicates in the common setting and 96% in the rare one (Monte Carlo SE about 4.6% and 2.8%). They cover $\beta_0 + w(s)$ for 94.8% and 94.3% of cells. Ignoring the uncertainty in $\boldsymbol{\theta}$ therefore costs little here, including at the prevalence of the data. This holds for the coefficient of a covariate that is independent in space and time and independent of w, which is the easiest case. S0 does not check spatially smooth covariates that are correlated with the field, and S3 shows that the plug-in standard error of δ is too small when a space-time field is fitted. A fit takes 2–4 seconds on one core.

4.3. *S1: residual structure after boosting.* Data are generated from (4.1) with the default parameters. All models are trained on the first 104 weeks and evaluated on the following 52 weeks at the same cells, as in the original analysis. We compare the true probabilities (oracle); the original-like XGBoost; the same model with each cell's training event rate as an additional feature, computed leaving out the row's own 52-week period (the location-aware baseline of Section 3.9); the unweighted XGBoost cross-fitted by country; the hybrid model (3.5) with independent cell effects and with the Matérn field; the parametric model (3.1) with country effects and field; and the persistence baseline. The parametric model has a single temperature slope, so it omits the biome-specific slopes of the truth and is not correctly specified. Table 3 reports out-of-sample scores over 20 replicates. Table S3 of the Supplementary Material reports Moran's I of the training-period cell residuals of the two XGBoost models and the recovery of the field. Figure 2 shows one replicate.

TABLE 2

S0: accuracy of the finite-volume SPDE field and of the empirical Bayes Laplace approximation, with fields drawn from the continuous Matérn process (50 replicates per setting). Mean over replicates, with the SD across replicates in parentheses, except for the coverage of b_1, where it is the Monte Carlo standard error.

Quantity	common ($b_0 = -2$, 50 reps)	rare ($b_0 = -4.5$, 50 reps)
Prevalence of event weeks	0.161 (0.030)	0.019 (0.005)
$\hat{\rho}_w$ (km) [truth 400]	420 (40)	409 (51)
$\hat{\sigma}_w$ [truth 1]	0.884 (0.064)	0.875 (0.075)
$\hat{b}_1$ [truth 0.5]	0.494 (0.016)	0.502 (0.021)
Coverage of 95% interval for b_1	0.88 (0.05)	0.96 (0.03)
Coverage of 95% intervals for $b_0 + w(s)$ (share of cells)	0.948 (0.009)	0.943 (0.017)
$\mathrm{cor}(\hat{w}, w)$	0.954 (0.008)	0.914 (0.018)
Fit time (s)	2.4 (0.7)	3.4 (0.8)
Marginal variance of w, interior cells [nominal 1]	1.07	–
Marginal variance of w, edge cells	2.02	–

Three findings stand out. First, boosting leaves strong spatial structure in its residuals. Moran's I of the training-period cell residuals is 0.36 (SD 0.05 across replicates) for the original-like XGBoost, which is trained on the same locations, and 0.64 for the version cross-fitted by country. Under independence it is about -0.001. The trees recover part of the field through country-identifying national covariates, so in-sample residuals understate the spatial structure. The cross-fitted offset alone predicts poorly (AUC-PR 0.10 against 0.17), as it must, since it ignores everything that is specific to a country.

Second, the field estimated on top of the cross-fitted offset recovers the truth (correlation between $\hat{w}$ and w 0.85), with $\hat{\rho}_w = 409$ km against 400 and $\hat{\sigma}_w = 0.83$ against 1.

Third, modelling the latent structure improves forecasts for the same places. Differences below are paired by replicate, with standard errors over the 20 replicates.

- *Against the original-like XGBoost.* The hybrid model with the field improves the log score in all 20 replicates (mean difference -0.0027, SE 0.0002) and AUC-PR in all 20 ($+0.011$, SE 0.001).
- *Field against independent cell effects.* The field beats independent cell effects in every replicate (log score -0.0013, SE 0.0001; AUC-PR $+0.003$, SE 0.0003): smoothing across neighbouring cells helps.
- *Against the location-aware baseline.* The location-aware boosting baseline is a strong competitor. It improves on the original-like XGBoost about as much as the hybrid model does. Against it, the hybrid model with the field has a better log score in 16 of 20 replicates (mean difference -0.0011, SE 0.0004) and a better AUC-ROC in 18 of 20 ($+0.008$, SE 0.001). Its Brier score is the same (difference $+0.00002$, SE 0.00006; better in 9 of 20) and its AUC-PR is similar (-0.001, SE 0.001; better in 6 of 20). The log-score gain closes about a quarter of the gap between the baseline and the true probabilities (0.0042). The hybrid model with independent cell effects is worse than the baseline: its log score is similar ($+0.0002$, SE 0.0004; better in 6 of 20) and its AUC-PR is lower (-0.004, SE 0.001; lower in 17 of 20).
- *Parametric model.* Although it omits the biome-specific temperature slopes, the parametric model beats the baseline in log score in every replicate (-0.0020, SE 0.0004) and in AUC-PR in 19 of 20 ($+0.004$, SE 0.001). It closes about half of the log-score gap to the true probabilities.

Once boosting is given each cell's own history, the gain from the latent field is therefore small. It appears in the log score and in AUC-ROC, but not in the Brier score or in AUC-PR, so

TABLE 3

S1: out-of-sample scores on the 52 test weeks (mean over 20 replicates, Monte Carlo standard error in parentheses). Lower is better for the log and Brier scores. The last two columns give the number of replicates in which the model has a better log score than (a) the original-like XGBoost and (b) the location-aware XGBoost with the cell training rate.

Model	Log score	Brier	AUC-PR	AUC-ROC	< (a)	<
True probabilities	0.1343 (0.0070)	0.0334 (0.0022)	0.200 (0.015)	0.805 (0.004)	20/20	20
XGBoost as in the original analysis	0.1401 (0.0072)	0.0341 (0.0023)	0.172 (0.015)	0.767 (0.006)	–	2/
XGBoost + cell training rate	0.1385 (0.0070)	0.0338 (0.0022)	0.185 (0.015)	0.781 (0.005)	18/20	
XGBoost, cross-fitted by country	0.1537 (0.0085)	0.0360 (0.0026)	0.101 (0.011)	0.670 (0.008)	0/20	0/
Cross-fitted XGBoost + u + iid cell effects	0.1387 (0.0071)	0.0339 (0.0023)	0.181 (0.015)	0.778 (0.005)	19/20	6/
Cross-fitted XGBoost + $u + w$	0.1374 (0.0071)	0.0339 (0.0023)	0.184 (0.015)	0.789 (0.005)	20/20	16
Parametric model with $u + w$	0.1365 (0.0071)	0.0336 (0.0022)	0.189 (0.015)	0.792 (0.005)	20/20	20
Persistence baseline	–	–	0.095 (0.010)	–	–	

these results do not show whether it comes from calibration or from ranking. The persistence baseline reaches an AUC-PR of 0.095, against 0.20 for the true probabilities. In the data, persistence is much stronger, because some cells record events almost every week. S1 does not reproduce that regime, which is one reason to repeat it with the calibrated parameters. With those parameters, [TODO: S1 calibrated: paired differences of hybrid_field, hybrid_iid and glmm against xgb_te (log score, Brier, AUC-PR, AUC-ROC), prevalence and persistence AUC-PR, from results/s1_paired_calibrated.csv and results/s1_scores_calibrated.csv].

4.4. *S2: national indicators under spatial confounding.* Three scenarios with 25 countries and 30 replicates each are considered: no confounding with $\alpha = 0$ ($\varrho = 0$); confounding with $\alpha = 0$ ($\varrho = 0.8$); and confounding with a true effect $\alpha = -0.3$. We compare estimators (i)–(iv) of Section 3.6, with spatial+ using $m = 10$ eigenvectors, and the XGBoost summary of the original analysis. For the latter, the original-like XGBoost is fitted to a fixed 30% subsample of the training data, as in the original procedure. The effect of Z_0 is summarised by the least-squares slope of its SHAP values on Z_0. Its intervals come from 40 replicates of a spatial block bootstrap with 2.5° blocks (48 blocks on the domain), each refitted with the same number of trees and class weight. The reported estimate is the median over bootstrap replicates, as in the original analysis. A single fit to the same subsample gives nearly the same estimates (means -0.01, 0.34 and 0.15 in the three scenarios below, against bootstrap medians of -0.01, 0.33 and 0.14). The estimate from a single fit to all training rows is [TODO: S2: mean est_full by scenario from results/s2_summary.csv (mean_est_full) after the rerun].

Figure 3 summarises the results (Table S4 of the Supplementary Material gives the numbers). Without confounding, all estimators are centred on zero, but intervals that ignore the country-level structure of the covariate are badly anticonservative. Logistic regression reports a standard error of 0.016 against an empirical SD of 0.107. It covers in 23% of replicates and declares a non-existent effect significant in 77%. Country effects alone restore nominal coverage. The block-bootstrap summary of the original analysis is closer to nominal than logistic regression, but its 95% interval still excludes zero in 20% of replicates (6 of 30; Monte Carlo SE 0.07). Its bootstrap standard error is about 0.70 times the empirical SD (0.044 against 0.063). Its interquartile range, the criterion used in the original analysis, excludes zero in 50% of replicates although Z_0 has no effect. This is the rate expected by construction: the interquartile range is a 50% interval, so as a test it has a nominal false-positive rate of 50%. It is not a significance criterion.

Under confounding with $\alpha = 0$, no estimator is unbiased:

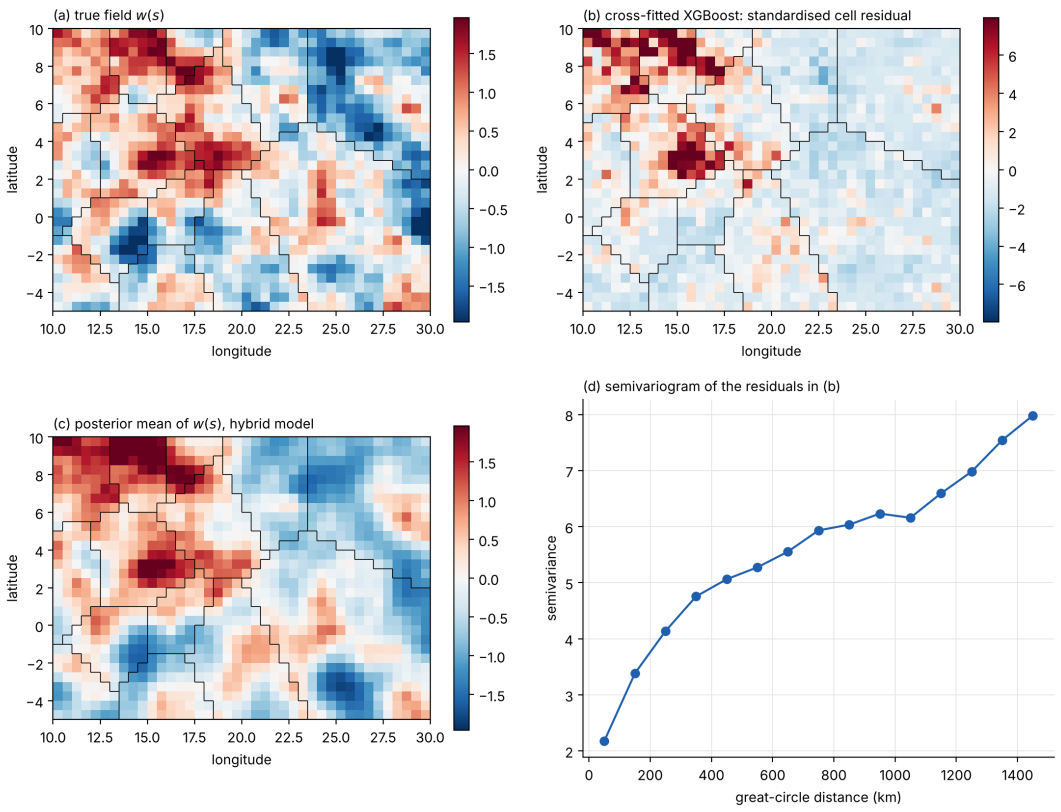


FIG 2. *S1, one replicate. From left: the true field w; standardised cell residuals of the XGBoost model cross-fitted by country (training period); posterior mean of w in the hybrid model (3.5); semivariogram of the XGBoost cell residuals. Colour scales are symmetric and capped at the 98th percentile of the absolute values. Lines are country borders.*

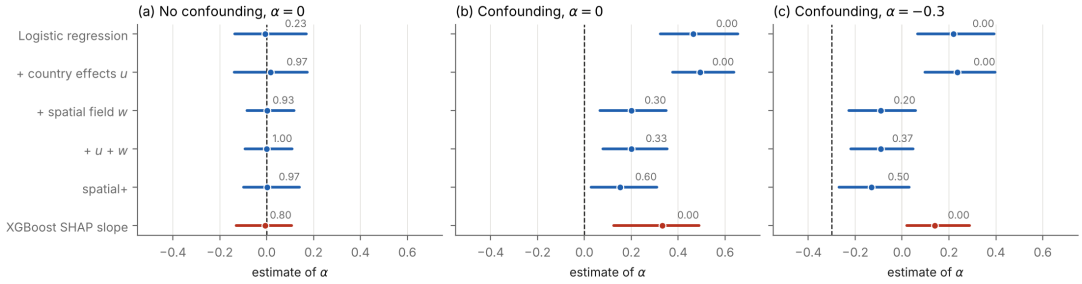


FIG 3. *S2: estimates of the effect α of a national indicator over 30 replicates per scenario. Dots are means and lines span the 5th to 95th percentiles of the estimates across replicates; the number above each line is the coverage of the nominal 95% intervals. Blue: latent Gaussian models; red: the XGBoost summary of the original analysis (least-squares slope of the SHAP values, block-bootstrap interval). The dashed line marks the true value. Full results are in Table S4 of the Supplementary Material.*

- logistic regression and the country-effects model report $\hat{\alpha}$ of 0.47 and 0.49 and always reject;
- the spatial field removes about 60% of the bias ($\hat{\alpha} = 0.20$), and spatial+ removes about 70% (0.15, coverage 0.60);

- the spatially adjusted estimators still reject $\alpha = 0$ in 70% (field), 67% (field and country effects) and 40% (spatial+) of replicates (Monte Carlo SE about 0.09);
- XGBoost reports a SHAP slope of 0.33, with an interquartile range that excludes zero in every replicate.

With a true protective effect ($\alpha = -0.3$), logistic regression, the country-effects model and XGBoost report effects of the *wrong sign* ($+0.22$, $+0.24$ and $+0.14$). The XGBoost interquartile range excludes zero in 97% of replicates. The spatial estimators recover the sign on average but only 30–45% of the magnitude (-0.09 to -0.13).

The lesson for the national sustainability indicators is that spatial adjustment reduces confounding bias but does not remove it. Under a confounded null, the adjusted estimators still reject in 40–70% of replicates, so an association that survives spatial adjustment is not evidence of an effect, and the data do not identify its causal size. A large SHAP contribution of a national indicator, stable across block-bootstrap replicates, is fully compatible with no effect at all. These results use a single confounder that varies at the country scale.

The block bootstrap resamples spatial blocks, but Z_0 varies only between countries, and a country spans several blocks. The bootstrap therefore does not reproduce the sampling variability of a country-level effect, whose effective sample size is the number of countries (Section 3.6). This is consistent with the bootstrap standard error being about 30% too small in the scenario without confounding (with 30 replicates, the empirical SD has a relative Monte Carlo error of about 13%).

4.5. *S3: state dependence versus persistent heterogeneity.* Three data-generating processes on a smaller 16×14 domain (224 cells, 6 countries, 104 training weeks, 20 replicates each) produce autocorrelation in Y_{st} for different reasons:

- *state dependence*: $\delta = 1$ and a weak field ($\sigma_w = 0.3$);
- *heterogeneity*: no state dependence ($\delta = 0$), a strong static field ($\sigma_w = 1.3$) and a space-time AR(1) field ($\sigma_v = 0.7$, $\phi = 0.9$ per week, $\rho_v = 300$ km);
- *mixed*: $\delta = 0.5$, $\sigma_w = 1$ and a space-time field with $\sigma_v = 0.5$ and $\phi = 0.9$.

Two further truths repeat the heterogeneity and mixed scenarios with a more persistent space-time field ($\phi = 0.97$ per week). We fit model (3.1) with lags only, with lags and w, and with lags, w and v. The space-time field v is resolved at weekly steps ($\Delta = 1$, the resolution of the data-generating process), at four-week steps ($\Delta = 4$) and at eight-week steps ($\Delta = 8$, the resolution used in the application). We also report the share of total mean absolute SHAP value that the unweighted XGBoost (300 trees, not the original procedure) assigns to the lag feature. Results are in Table 4.

Under state dependence, ignoring the latent field inflates $\hat{\delta}$ by about a quarter (1.26; coverage 0.60), and all latent specifications are approximately unbiased (0.99–1.03) with coverage of 0.90–0.95. Under heterogeneity the contrast is much sharper. With lags only, $\hat{\delta} = 1.61$ although the true value is zero. Adding the static field reduces the estimate to 0.46, but no interval covers zero: slowly varying episodes act as state dependence. Only the models with the space-time field bring the estimate close to zero: $\hat{\delta}$ is 0.05 with four-week steps and 0.01 with weekly steps, both within Monte Carlo error of zero (Monte Carlo SE about 0.035). The mixed scenario behaves in the same way. The static field leaves an upward bias (0.78 against 0.5), which the space-time field removes (0.55 and 0.54, Monte Carlo SE about 0.04).

The standard errors of $\hat{\delta}$ are too small when v is fitted. In the heterogeneity and mixed scenarios the empirical SD exceeds the mean standard error by 27–43% (ratios 1.27–1.43), and coverage is 0.80–0.90 (Monte Carlo SE 0.07–0.09). In the state-dependence scenario, where the truth has no v, the ratio is 0.94–1.02 for the three latent models. With 20 replicates each ratio has a Monte Carlo SE of about 0.2, so the size of the shortfall is imprecise, but it

TABLE 4

S3: estimation of the state-dependence parameter δ (20 replicates per scenario), and the share of the XGBoost mean absolute SHAP value assigned to the lag feature.

Estimator	Mean $\hat{\delta}$	Emp. SD	Mean SE	Coverage
State dependence ($\delta = 1$)				
Lags only	1.263	0.107	0.136	0.60
Lags + w	1.032	0.131	0.139	0.90
Lags + w + v (4-week steps)	0.990	0.135	0.140	0.95
Lags + w + v (weekly steps)	0.997	0.143	0.140	0.95
XGBoost: share of mean |SHAP| for lags	0.108	0.034	–	–
Heterogeneity ($\delta = 0$)				
Lags only	1.614	0.386	0.099	0.00
Lags + w	0.464	0.093	0.111	0.00
Lags + w + v (4-week steps)	0.045	0.152	0.119	0.90
Lags + w + v (weekly steps)	0.011	0.162	0.120	0.90
XGBoost: share of mean |SHAP| for lags	0.094	0.038	–	–
Mixed ($\delta = 0.5$)				
Lags only	1.659	0.351	0.106	0.00
Lags + w	0.782	0.111	0.115	0.25
Lags + w + v (4-week steps)	0.545	0.152	0.120	0.85
Lags + w + v (weekly steps)	0.535	0.172	0.120	0.80
XGBoost: share of mean |SHAP| for lags	0.116	0.044	–	–

appears in all four cells with v. Intervals for δ from the space-time model should therefore be read as somewhat too narrow.

Resolving v at four-week steps instead of weekly steps changes $\hat{\delta}$ little in these scenarios, and it is four to five times faster. Paired by replicate, the four-week estimate is larger by 0.034 (SE 0.006) under heterogeneity and by 0.010 (SE 0.008) in the mixed scenario. The space-time field has an AR(1) coefficient of 0.9 per week, so it changes slowly relative to the four-week history window. This argument is weaker at the eight-week steps used in the application, since $0.9^8 \approx 0.43$. With $\Delta = 8$, [TODO: S3: mean, emp. SD, mean SE and coverage of $\hat{\delta}$ for lag+field+st8 in all five truths, and the $\phi = 0.97$ rows, from results/s3_summary.csv after the rerun]. Faster space-time dynamics ($\phi < 0.9$) are not covered.

Finally, the XGBoost importance of the lag feature is about the same under all three truths (share 0.09–0.12, Monte Carlo SE about 0.01). With nine features, this is close to an equal share, so these simulations do not reproduce the dominance of recent history seen in the data. The share is also normalised by the total, which changes with the strength of the field. The simulations show only that the same importance of recent history can arise with or without state dependence. Importance alone therefore cannot show that violence begets violence.

4.6. *S4: biome-specific climate effects.* The temperature effect follows (3.4), with biome means $\boldsymbol{\gamma} = (-0.3, -0.1, 0.1, 0.3)$ and a within-biome Matérn deviation η ($\sigma_\eta = 0.15$, $\rho_\eta = 600$ km), $\sigma_w = 0.8$ and 30 replicates. We compare:

- fixed biome-specific slopes, without and with $u + w$;
- the SVC model (3.4);
- two XGBoost summaries fitted to the anomaly: the per-biome least-squares slope of the SHAP value of the anomaly on the anomaly, and the same slope computed from the SHAP value plus the interaction values credited to the anomaly ($\Phi_{jj} + 2\sum_{k \neq j} \Phi_{jk}$ in the notation of Lundberg et al., 2020). For a pure biome-by-temperature interaction $\gamma_b z$ with biome independent of z, the interventional SHAP value of z is $(\bar{\gamma} + \gamma_b)(z - \mathbb{E}z)/2$, where $\bar{\gamma}$ is

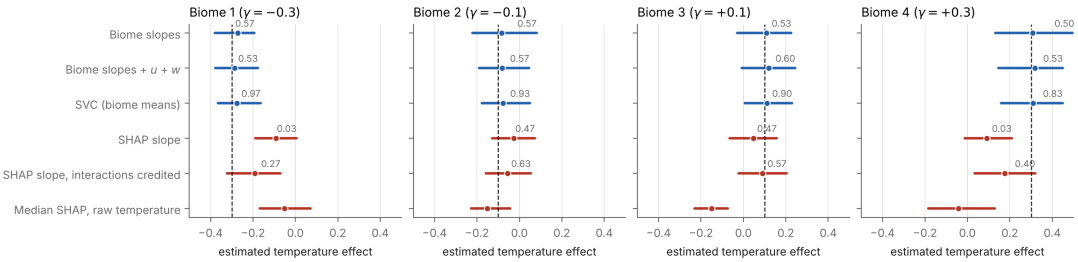


FIG 4. *S4: estimates of the biome-specific temperature effects over 30 replicates. Dots are means and lines span the 5th to 95th percentiles of the estimates across replicates; the number above each line is the coverage of the nominal 95% intervals for γ_b (not shown for the median SHAP value, which has no corresponding estimand). Blue: latent Gaussian models; red: XGBoost summaries. The dashed line marks γ_b. Full results are in Tables S5 and S6 of the Supplementary Material.*

the cell-weighted mean of γ_b. The first summary therefore targets $(\bar{\gamma}+\gamma_b)/2$, which equals $\gamma_b/2$ here only because the biomes have equal sizes and $\boldsymbol{\gamma}$ is symmetric about zero. The second targets γ_b in general. A further run with $\boldsymbol{\gamma} = (0, 0.1, 0.2, 0.4)$ checks this;

- the statistic of the original analysis: the median SHAP value of raw temperature within each biome, from a model fitted to raw temperature.

The XGBoost models are original-like. As in the original procedure, each estimate is the median over 20 replicates of the 2.5° block bootstrap on a fixed 30% subsample, and the 95% intervals are bootstrap percentiles. With 20 replicates these are close to the range of the replicates. For the median SHAP value we report the interquartile range of the bootstrap replicates, the criterion of the original analysis. We also report the same statistics from a single fit to all training rows. Coverage is reported for both the biome mean γ_b and the realised biome average $\bar{\beta}_b$ (Section 3.5).

The SVC model is the only estimator with close to nominal coverage for the biome means γ_b (0.83–0.97, Monte Carlo SE 0.03–0.07; Figure 4 and Table S5 of the Supplementary Material), although the value of 0.83 in biome 4 is about 1.7 Monte Carlo SE below 0.95. Its intervals are conservative for the realised biome averages $\bar{\beta}_b$ (coverage 1.00), as expected, because they also account for the sampling of regions within biomes. Fixed biome slopes are nearly unbiased, but their intervals ignore heterogeneity within biomes. They cover γ_b in only 50–60% of replicates (Monte Carlo SE about 0.09), and $\bar{\beta}_b$ in 53–90%. The price of honest intervals is power: the SVC model detects the small effects ($|\gamma_b| = 0.1$) in only 23–33% of replicates. The SVC model also recovers the coefficient surface (Table S6 of the Supplementary Material). The correlation between $\hat{\beta}(s)$ and $\beta(s)$ is 0.92, the RMSE is 28% lower than with biome slopes only (0.106 against 0.147), and $\hat{\sigma}_\eta = 0.13$ against a true 0.15.

The XGBoost summaries are much less informative:

- *SHAP slopes.* The per-biome slopes of the ordinary SHAP values have the right ordering but are between 0.26 and 0.48 times γ_b (about 0.3 in three of the four biomes), below the $\gamma_b/2$ expected in this design from the splitting of interactions. Crediting the interaction values to temperature raises them to between 0.56 and 0.89 times γ_b. The remaining attenuation may come from this configuration of boosting (a 30% subsample of about 37,000 rows, a minimum child weight of 30 and early stopping) rather than from boosting in general. We have not separated these causes. With a single fit to all training rows, [TODO: S4: mean est_full / truth for xgb_slope and xgb_slope_full by biome, from results/s4_summary.csv after the rerun]. With the asymmetric $\boldsymbol{\gamma}$, [TODO: S4 asym: mean xgb_slope against shap_target and against γ_b, and mean xgb_slope_full against γ_b, from results/s4_summary_asym.csv].

- *Median SHAP values.* The median SHAP value of raw temperature within a biome, the statistic of the original analysis, is negative in *every* biome (-0.04 to -0.15), whatever the true effect. Its bootstrap interquartile range excludes zero with a negative sign in every replicate in biome 3 and in 53% of replicates in biome 4, where the true effects are positive. Applying the original criterion would therefore report a "negative contribution of temperature" where the effect is positive.

As argued in Section 3.5, the median SHAP value describes where a biome's covariate values lie relative to the model baseline, not the strength of the effect.

5. Application to ACLED 2018–2024. All models are trained on 2018–2022 and evaluated on 2023–2024. The boosting models use the hyperparameters of Menicali, Marcantonio and Castruccio (2026). The test period includes the onset of new large wars, in Sudan from April 2023 and in Gaza from October 2023, which a model trained on 2018–2022 cannot anticipate from recent history. Besides the scores on all test rows, we therefore report scores on onset rows: test cell-weeks in cells with no event of that outcome in the previous 52 weeks (Section 3.9). The computations were run at the Notre Dame Center for Research Computing.

5.1. *Reconstructing the cell-week table.* The table built by the original pipeline has 70,033,444 rows but only 34,671,180 distinct cell-weeks (Section 2.1). In one partition, 613,459 cell-weeks appear once, while others appear 2, 3, 8, 16, 32, 48 and up to 648 times. The national indicators and IPC regions are attached by spatial intersection and then merged on cell and year. A cell that touches several countries or IPC regions is therefore multiplied at every join, and each copy combines indicators from different units.

The duplication matters for more than the covariates. The four-week event counts of the original pipeline are computed row by row within each cell, after sorting by cell and week. When a cell-week has copies, the rows immediately before a given row are copies of the same week, so the "past" count contains the current week's events, which are the outcome. We recomputed the original counts exactly and compared them with the true counts over the four previous weeks, on 69.4 million rows of the duplicated table, over all years and without the coverage filter. Among rows with an event, counting each copy separately, 22% have an original count that includes the current week's events while the true past count is zero, for both outcomes. Used directly as a predictor of the outcome on these rows, without any model, the original count reaches an AUC-PR of 0.883 for conflict and 0.833 for unrest. The true count reaches 0.495 and 0.450. These values describe the duplicated table as a whole. They are not on the same rows as the test-period scores below. The leak arises only in cell-weeks that have copies. Copies come from cells that intersect several countries or several IPC regions; we have not checked how many of them lie on international borders.

The repaired table has one row per cell-week (Section 2.1). After ACLED coverage and the biome exclusion it has 19,365,684 cell-weeks in 59,344 cells and 366 weeks. The test period has [TODO: x] cell-weeks in [TODO: x] cells with complete climate records.

5.2. *The boosting analysis before and after the repair.* Table 5 reports three versions of the XGBoost analysis. The first, *as published*, applies the original procedure to the duplicated table: the climate variables of the current week and their four-week means as features, positives weighted by $(n_0/n_1)^{1/2}$, 5 million training rows sampled with replacement, and the number of trees chosen by early stopping on the test years. The second, *deduplicated*, applies the same procedure unchanged to the repaired table. It isolates the effect of the duplication and the leak. The third, *corrected*, also changes the procedure. The number of trees is chosen by early stopping on 2022 after training on 2018–2021, and the model is then refitted on 2018–2022. It uses no class weight. Its features use only information available before

TABLE 5

XGBoost on the 2023–2024 test period. As published: duplicated table, original procedure (current-week features, class weight, early stopping on the test years). Deduplicated: repaired table, original procedure. Corrected: repaired table, features available before the week, no class weight, early stopping on 2022. Corrected + cell event rate: the location-aware model. AUC-PR is also reported separately for cell-weeks that have copies in the original table and for the others. The persistence baseline ranks cell-weeks by the number of events in the previous four weeks. The as-published rows are on a different test set from the other rows, with a different prevalence.

Pipeline	Outcome	Test rows	Prevalence	Trees	AUC-PR	AUC-PR by cell		Persiste
						dup. in original	other cells	AUC-I
As published	Conflict	14.43M	2.98%	1998	0.934	0.966	0.498	0.89
As published	Unrest	14.43M	3.58%	1474	0.917	0.968	0.528	0.85
Corrected	Conflict	4.79M	1.62%	810	0.544	0.588	0.522	0.51
Corrected	Unrest	4.79M	2.73%	879	0.557	0.547	0.559	0.50
Corrected + cell event rate	Conflict	4.79M	1.62%	195	0.564	0.598	0.546	0.51
Corrected + cell event rate	Unrest	4.79M	2.73%	147	0.583	0.567	0.586	0.50

the week, so the climate variables of the current week are dropped and only their four-week means are kept. The 5 million training rows are sampled without replacement. The weighted models are prior-corrected before scoring (Section 3.7). We also fit the location-aware model of Section 3.9, which adds each cell's training event rate to the corrected model. For the fit that chooses the number of trees, this rate uses 2018–2021 only; for the final fit it leaves out the row's own year.

On the duplicated table we recover the published performance, with AUC-PR [TODO: x] for conflict and [TODO: x] for unrest, against 0.93 and 0.91 reported. Split by cell, nearly all of it comes from the cell-weeks that have copies in the original table (AUC-PR [TODO: x]), and on the other cell-weeks AUC-PR is [TODO: x]. The first two rows of Table 5 are on different test sets: the duplicated table has [TODO: x] test rows with conflict prevalence [TODO: x]%, the repaired table [TODO: x] rows with [TODO: x]%. AUC-PR depends on prevalence, so part of the difference between them is not due to the model. The deduplicated and corrected rows are on identical test rows and can be compared directly. Deduplication alone gives AUC-PR [TODO: x] for conflict and [TODO: x] for unrest. The corrected model reaches [TODO: x] and [TODO: x], the persistence baseline [TODO: x] and [TODO: x], and the location-aware model [TODO: x] and [TODO: x]. On onset rows the corrected model reaches [TODO: x] and [TODO: x]. [TODO: Interpretation: how much the covariates add to recent history, how much a cell intercept adds to the covariates, and how much survives on onset rows.]

5.3. *Residual spatial structure and the hybrid model.* We fit the hybrid model (3.5) to Africa and the Middle East (longitude 20°W to 65°E, latitude 36°S to 45°N; [TODO: x] cells). It combines the XGBoost offset cross-fitted by country, country effects, and either a Matérn field with sea barriers (Remark 1) or independent cell effects. The offset is trained on the rows of this region only, whereas the corrected XGBoost model of Section 5.2 is trained on all locations worldwide. The comparisons below are on identical test rows, but the training sets differ.

Table S7 of the Supplementary Material reports the test-period residual diagnostics of every model, and Figure S1 of the Supplementary Material the semivariograms of the test-period residuals. [TODO: Residual structure of the corrected XGBoost model (Moran's I, within-country and across-border correlations, semivariogram) for conflict and unrest, and

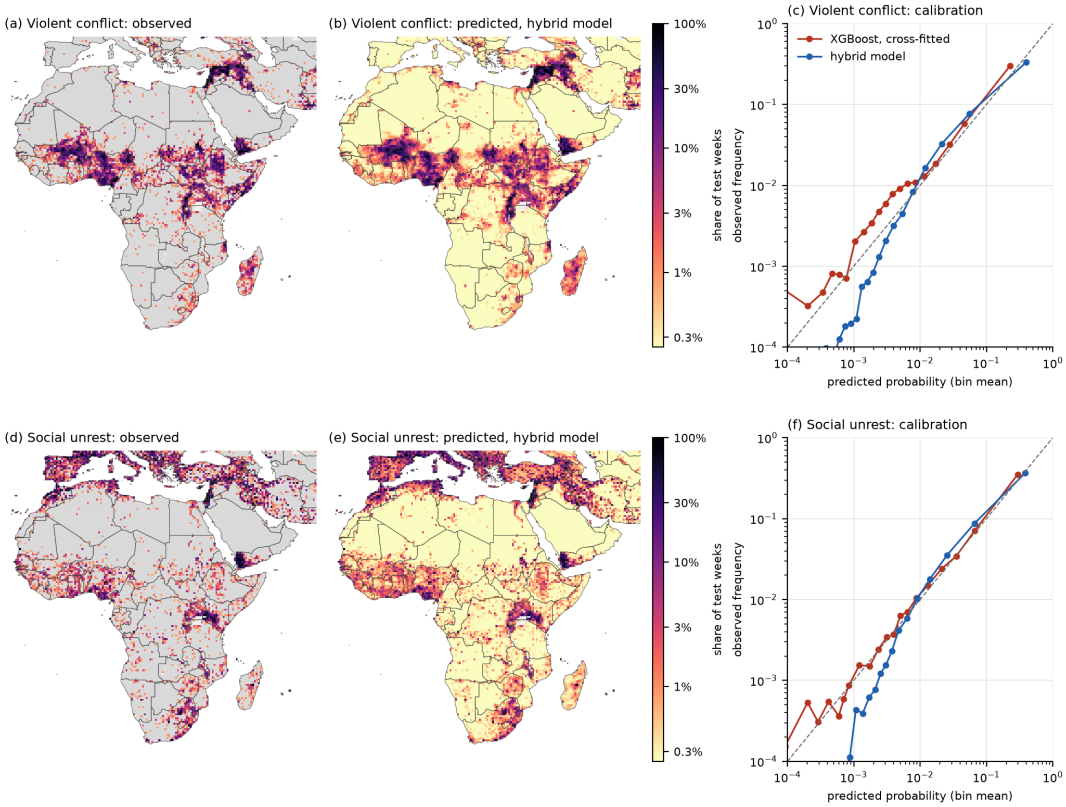


FIG 5. *Observed and predicted event frequency in the test period 2023–2024, Africa and the Middle East. (a, d) Share of test weeks with at least one event in each cell; grey cells have no event week. (b, e) Mean predicted probability of the hybrid model with country effects and the Matérn field over the same weeks, on the same colour scale. (c, f) Calibration: observed frequency against mean predicted probability in 20 equal-count bins, for the hybrid model and the XGBoost offset cross-fitted by country; the dashed line is perfect calibration. [TODO: Regenerate from the Round 3 fits.]*

how much of it remains after the location-aware model and the hybrid models. For the cross-fitted offset, the across-border correlation split by same fold and different fold (Section 3.9).]

Table S8 of the Supplementary Material gives the fitted hyperparameters. [TODO: Range and standard deviation of the fields; whether the range is resolved by the 0.5° grid (Section 3.2).] A range close to the size of a cell, about 55 km at the equator, would mean that the field is not resolved by the grid. The field then behaves like independent cell effects, and it should not be read as regional hotspots that cross borders. [TODO: State whether this holds for unrest, and compare the field with the independent cell effects in Table 6.] For unrest, cell-level effects would largely track towns and cities, where protests are both more frequent and more often reported.

Figure 5 compares observed and predicted frequencies over the test period. The hybrid model reproduces the regional pattern of both outcomes, including the Sahel, eastern Congo, Somalia, Yemen and Syria–Iraq for conflict. [TODO: Describe calibration from the Round 3 fits: in the current fits the hybrid model is close to the diagonal above a probability of about 1% and overpredicts below it, while the cross-fitted offset underpredicts in the same bins.]

The estimated fields are shown in Figure 6. Table S9 of the Supplementary Material summarises the posterior of the field averaged over eleven regions. The regions were chosen before the fit: eight known conflict zones and three regions expected to be quiet. A posterior probability close to one that the average over a known conflict zone is positive is therefore

TABLE 6

Out-of-sample scores, Africa and the Middle East, 2023–2024, on identical test rows. Lower is better for the log and Brier scores. Differences in mean log score against the corrected XGBoost model, with standard errors clustered by cell and by country. The hybrid offset is trained on Africa and the Middle East only; the corrected and location-aware XGBoost models are trained on all locations.

Outcome	Model	Log score	Brier	AUC-PR	AUC-ROC	Δ log score $\times 10^3$ (S
						by cell
Conflict	XGBoost, corrected	0.0614	0.0160	0.514	0.951	reference
Conflict	XGBoost + cell event rate	0.0585	0.0156	0.534	0.959	-2.94 (0.24)
Conflict	XGBoost, cross-fitted by country	0.0716	0.0175	0.444	0.916	10.21 (0.47)
Conflict	Cross-fitted XGBoost + u + iid cell effects	0.0609	0.0163	0.521	0.958	-0.54 (0.17)
Conflict	Cross-fitted XGBoost + $u + w$	0.0605	0.0163	0.521	0.959	-0.88 (0.18)
Conflict	Persistence (past 4 weeks)	–	–	0.446	–	–
Unrest	XGBoost, corrected	0.0658	0.0172	0.536	0.950	reference
Unrest	XGBoost + cell event rate	0.0627	0.0169	0.550	0.960	-3.04 (0.19)
Unrest	XGBoost, cross-fitted by country	0.0697	0.0179	0.507	0.939	3.93 (0.22)
Unrest	Cross-fitted XGBoost + u + iid cell effects	0.0634	0.0170	0.547	0.959	-2.35 (0.15)
Unrest	Cross-fitted XGBoost + $u + w$	0.0634	0.0170	0.547	0.959	-2.37 (0.16)
Unrest	Persistence (past 4 weeks)	–	–	0.451	–	–

expected and is not a finding in itself. The last two rows of the table give a summary that does not depend on chosen regions: the share of cells whose posterior probability that $w(s) > 0$ exceeds 0.95, and the share for which it is below 0.05. [TODO: Shares of cells for conflict and unrest; where the cells with clearly positive and clearly negative w are.] Negative values of w mark low *recorded* risk relative to the covariates, recent history and country effects. In regions with restricted media access and few ACLED sources, such as the interior of the Arabian peninsula, they may reflect reporting rather than occurrence (Section 2.1). Positive values in cells with towns or administrative centres may in part reflect events of low spatial precision, which ACLED places at such centroids. Because the offset is cross-fitted by country, errors of the offset that are common to a country are absorbed by u, but errors that vary within a country can be absorbed by w.

Table 6 compares out-of-sample scores on identical test rows: the corrected XGBoost model, the location-aware model, the cross-fitted offset alone, the hybrid model with independent cell effects, the hybrid model with the field, and the persistence baseline. All are computed with the same code, on all test rows and on onset rows. Differences in log score and Brier score are given against the corrected XGBoost model, with standard errors clustered by cell and by country. A gain of a hybrid model over the cross-fitted offset alone is not evidence for the latent effects (Section 3.7). The relevant comparisons are with the corrected and location-aware models, and between the field and the independent cell effects. [TODO: Results: hybrid with the field against hybrid with independent cell effects (does the spatial smoothing matter beyond a cell intercept?), against the location-aware boosting model, and against persistence, on all rows and on onset rows. Global fit.]

5.4. *State dependence, climate and national indicators.* [TODO: Results of the parametric models, all with independent cell effects in addition to the terms listed: (i) the state-dependence coefficient δ with lags only, with $u + w$, and with the space-time field v in the central Sahel, read as conditional dependence of recorded events (Section 3.4); (ii) biome means and within-biome variation of the maximum-temperature effect from the SVC model, against fixed biome slopes; (iii) the between- and within-country coefficients of the national indicators without latent effects, with u, with $u + w$, and with spatial+ for $m = 10$, 20 and

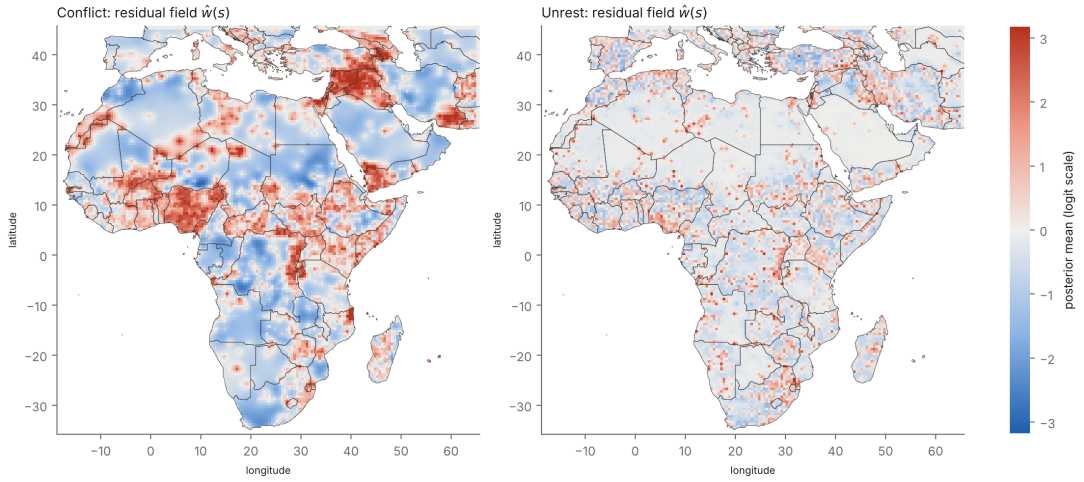


FIG 6. *Posterior mean of the residual field $w(s)$ of the hybrid model for conflict (left) and unrest (right), on the logit scale. Positive values mark higher recorded risk than predicted by the covariates, recent event history and country effects. Recorded risk mixes occurrence and reporting, and events of low spatial precision are placed at town and administrative centroids (Section 2.1).*

m^*, read in the light of Section 4.4; (iv) the coefficients of the missingness indicators, and the national-indicator coefficients refitted on the countries with complete indicators.]

6. Discussion.

[TODO: Discussion (not drafted yet).]

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SUPPLEMENTARY MATERIAL

Code

Python implementation of the latent Gaussian model, the simulator and all simulation studies.

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